

NAVAL FIGHTER WEAPONS SCHOOL

FLIGHT OPERATIONS MANUAL

TRAINING SQUADRON EDITION

Radio Procedures • Pre-Flight Checklists • Callsign Protocol • Weather Minimums

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FOR TRAINING AND SIMULATION USE ONLY — NOT FOR OPERATIONAL FLIGHT

“Speed is life. Altitude is life insurance. A checklist is how we keep both from becoming paperwork.”

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1. Radio Procedures

All radio communication follows standard aviation phraseology. Read back all clearances, altitudes, and headings. Keep transmissions short, clear, and in the correct sequence: who you're calling, who you are, and what you want. If your transmission needs a second cup of coffee to understand, it is too long.

1.1 Standard Call Structure

Every transmission follows the same three-part structure, because “hello?” is not a flight plan:

1. Station being called (“Operations, this is...”)
2. Your callsign (“...Viper One...”)
3. Your message, request, or report (“...requesting taxi, VFR to the training area.”)

1.2 Standard Phraseology

Phrase	Meaning
Affirm / Negative	Yes / No
Copy	Message received and understood
Say again	Repeat your last transmission
Standby	Wait — I will respond shortly
Wilco	Message understood, will comply
Roger	Message received (not necessarily agreement)
Checking in	Reporting on frequency, ready for instructions
Checking out	Leaving the frequency
Bingo	Minimum fuel to return to base
Knock it off	Stop the current maneuver/exercise immediately
Fox, Fox, Fox	Simulated weapon released, tone practice

1.3 Emergency & Priority Calls

- MAYDAY, MAYDAY, MAYDAY — grave and imminent danger, immediate assistance required.
- PAN-PAN, PAN-PAN, PAN-PAN — urgent situation, not immediately life-threatening.
- Declare fuel state as “Bingo” or “Joker” as soon as it applies — do not wait to be asked.
- Emergency calls take priority over all other radio traffic; all other stations hold transmissions.

1.4 Frequency Discipline

- Check in on the assigned frequency before any request or maneuver.
- Keep transmissions under 5 seconds where possible — think before you key the mic, because the whole squadron can hear your thought process if you do not.
- Never transmit over another aircraft's call; wait for a clear break.
- Read back all altitude, heading, and clearance instructions in full.

2. Pre-Flight Checklist Basics

Every sortie begins with a full pre-flight sequence. No aircraft launches without all four phases signed off: Walk-Around, Cockpit Setup, Systems Check, and Final Clearance. Skipping one is how harmless-looking little oversights become very expensive training aids.

2.1 Walk-Around Inspection

4. Confirm aircraft tail number matches the flight plan; borrowing the wrong jet is frowned upon in every department.
5. Inspect fuselage, wings, and control surfaces for visible damage.
6. Check landing gear, tires, and brakes for wear or leaks.
7. Verify all access panels are secured.
8. Confirm pitot tube and static ports are unobstructed and covers removed.
9. Check fuel and oil levels against the flight plan requirement.

2.2 Cockpit Setup

10. Verify ejection seat safety pins status matches phase (installed on ground, removed pre-taxi per SOP); this is not the system to “figure out later.”
11. Set altimeter to current field pressure.
12. Confirm avionics and navigation systems power up and self-test clean.
13. Set radios to assigned tower and squadron frequencies.
14. Verify fuel quantity indicator matches walk-around reading.

2.3 Systems Check

- Flight controls — free and correct movement in all axes; the aircraft should respond like a jet, not a shopping cart.
- Hydraulics — pressure in normal range, no warning lights.
- Engine instruments — all readings within normal operating range.
- Warning and caution panel — clear, no unresolved faults.
- Emergency oxygen — charged and accessible.

2.4 Final Clearance

- Confirm current weather is at or above minimums for the mission (see Section 4).
- Confirm flight plan is filed and active.
- Obtain taxi clearance from ground control before moving.
- Report “Ready for departure” only once all checklist items are complete. “Mostly ready” is not a clearance category.

3. Callsign Rules

Callsigns identify pilots on the radio and in all flight plan records. They are assigned, not self-selected, which is fortunate because otherwise every roster would include at least six Mavericks. Once assigned, callsigns are used exclusively in place of rank and name during flight operations.

3.1 Assignment

- Callsigns are assigned by the Squadron Commander after a pilot's first flight review, typically based on a notable incident, personality trait, or a play on the pilot's name. In other words, history may not remember everything, but the squadron will.
- A callsign is assigned once and used for the duration of a pilot's assignment to the squadron.
- Callsigns must be unique within an active squadron roster at all times.

3.2 Format in Radio Calls

Callsigns are used with a flight number when operating as part of a formation, and alone when operating as a single aircraft:

- Formation lead: "Viper One"
- Formation wingman: "Viper Two"
- Single-ship: "Maverick"

3.3 Format in Flight Plan Records

All flight plan records reference the pilot by callsign in the Pilot field, with legal name retained only in the underlying roster record for administrative purposes. This keeps radio traffic, briefings, and flight plan lookups consistent, and prevents paperwork from sounding like a family reunion.

3.4 Etiquette

- Never use a pilot's given name over the radio during an active mission.
- Do not assign or use an unofficial callsign in official records or radio traffic.
- Flight instructors and RIOs are addressed by callsign the same as pilots.

4. Weather Maximums

No sortie may be flown over the maximums for its mission type. The Pilot in Command confirms current conditions are less or equal to these values before requesting final clearance. The sky may be dramatic, but it does not get a vote over safety.

4.1 Mission Sortie Maximums

Mission Type	Rain Chance	Wind Speed	Temperature High
Training Exercise	50	9	40
Reconnaissance	0	10	75
Combat Simulation	0	10	32
Night Operations	100	10	80
Patrol Mission	75	15	80
Advanced Training	100	20	80
Interceptor Drill	100	25	100
Navigation Training	25	10	32
Carrier Landing Practice	25	15	75
Carrier Operations	0	20	60
Test Flight	0	9	40

4.2 No-Go Conditions

- Thunderstorms within 25 nm of the training area or planned route. If the sky is auditioning for a disaster movie, the sortie waits.
- Icing conditions forecast at any planned altitude.
- Snowfall predicted.
- Any active wind shear or microburst advisory for the departure or recovery field.

4.3 In-Flight Weather Changes

- If conditions rise above maximums for the mission type while airborne, the flight lead calls “knock it off” and initiates recovery. Weather does not improve just because everyone is having fun.
- Report deteriorating conditions to Operations immediately — do not wait for end of mission.

5. Quick Reference Card

A condensed summary for pilots and RIOs — for full detail, see the relevant section above. This is the “I have thirty seconds and one glove on” version.

Radio

- Structure: Who you're calling → Who you are → What you want. Radio is not the place for a mystery novel.
- Emergency: MAYDAY (grave danger) / PAN-PAN (urgent, non-critical).
- Declare fuel state (Bingo/Joker) proactively.

Pre-Flight

- Walk-Around → Cockpit Setup → Systems Check → Final Clearance.
- No “Ready for departure” call until all four phases are complete; optimism is not a checklist item.

Callsigns

- Assigned by the Squadron Commander — never self-selected, no matter how strongly someone feels about “Thunder Falcon.”
- Used in place of name on radio and in flight plan records.

Weather

- Basic VFR sortie: 3,000 ft ceiling / 5 sm visibility / 25 kt wind.
- Any mission: no-go on thunderstorms within 25 nm or unforecast icing; aircraft prefer not to be used as weather probes.

END OF MANUAL